

Supplementary Figure S5. Directed Acyclic Graph (DAG) for KDIGO AKI Study

Legend:

- Red box = Exposure (KDIGO creatinine-based AKI staging)
- Blue box = Outcome (ICU in-hospital mortality)
- Green ellipse = Measured confounders (adjusted in Model B)
- Orange ellipse (dashed) = Unmeasured confounders
- Yellow diamond = Collider variables (NOT adjusted; conditioning acknowledged in Limitations)
- Purple ellipse = Mediator (RRT; adjusted in Model B as part of severity)

Note: ICU admission and SCr measurement availability are structural colliders inherent to the MIMIC-IV design. Conditioning on ICU admission is unavoidable in ICU-based cohorts. SCr measurement availability affected 1,075 (1.3%) of 85,242 ICU stays. Neither was included as a covariate in the regression models. RRT initiation is both a component of KDIGO Stage 3 definition and a mediator on the AKI → mortality pathway; it was adjusted in Model B as a severity marker and further decomposed in stratified analyses.

